Maya Pal Gambhir

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Education

Princeton University — Aug. 2026 – Present
Princeton, NJ — *Ph.D. in Computer Science*
Advisors: Prof. Aleksandra Korolova and Prof. Jonathan Mayer

University of Pennsylvania — Aug. 2022 – May 2026
Philadelphia, PA — *B.S.E. in Artificial Intelligence* *GPA: 3.87/4.0*

Research Experience

Princeton University, Center for Information Technology Policy — Aug. 2026 – Present
Princeton, NJ — *Graduate Researcher*
Advisors: Prof. Aleksandra Korolova and Prof. Jonathan Mayer

- Studying the measurability of advertising effectiveness for small businesses using causal inference, with a focus on structural barriers that limit access to meaningful ad performance data.

University of Pennsylvania, Department of Computer Science — Jan. 2024 – May 2026
Philadelphia, PA — *Undergraduate Research Assistant*
Advisors: Prof. Aaron Roth, Prof. Surbhi Goel, and Prof. Bailey Flanigan (MIT)

- Developed conformal prediction methods for LLM reasoning chains with provable factuality guarantees; achieved 90% factuality retaining 80%+ of model outputs. Published at **ICLR 2025**.
- Designed a minimax optimization algorithm for dropout-robust citizens' assembly panel selection with provable fairness guarantees. Published at **AISTATS 2026** (Spotlight).
- Identified compact statistics predicting sortition algorithm performance to within 1–2% error and proposed a privacy-preserving synthetic data pipeline for open benchmarking (undergraduate thesis).

University of Pennsylvania, Computational Social Science Lab — Sept. 2023 – Jan 2024
Philadelphia, PA — *Undergraduate Research Assistant*
Supervisor: Baird Howland (Duncan Watts Lab)

- Evaluated LLM performance on emotion detection across a large news corpus; developed API prompting strategies for quote extraction and analyzed error rates.

Publications

Maya Pal Gambhir, Bailey Flanigan, Aaron Roth. "Near Optimal Dropout-Robust Sortition." *AISTATS*, 2026. **Spotlight.**

Maya Pal Gambhir[*], Maxon Rubin-Toles[*], Keshav Ramji, Aaron Roth, Surbhi Goel. "Conformal Language Model Reasoning with Coherent Factuality." *ICLR*, 2025. ([*]equal contribution)

Work Experience

Data Engineering Intern, Munich Re — Jun. 2023 – Aug. 2023
New York City, NY — *Software Engineering Intern*

- Built a React-based frontend for an internal insurance estimation API; created technical documentation and testing suite for production deployment.

Teaching Experience

Teaching Assistant, University of Pennsylvania

CIS 5230 (Graduate): Ethical Algorithm Design — *Spring 2026*
CIS 3333: Mathematics of Machine Learning — *Fall 2025*
CIS 1600: Mathematical Foundations of CS — *Summer 2025*

Awards and Honors

AI Academic Excellence Award, Raj and Neera Singh BSE in AI Program — 2026
Wolf-Hallac Award, Penn Engineering (outstanding graduating female senior) — 2026
CRA Undergraduate Research Award, Honorable Mention — 2025
Rachleff Scholars Program, University of Pennsylvania — 2022 – 2026

Technical Skills

Languages & Tools: Python, Java, SQL, JavaScript/TypeScript, PyTorch, JAX, scikit-learn, cvxpy, React